

4 (a) The magnetic flux through a surface is the product of the magnetic flux density perpendicular to the surface and the area of the surface.

(b) Magnetic flux linkage

$$= NBA = (500)(5.0 \times 10^{-2})(2.5 \times 10^{-2})$$

$$= 0.625 \text{ wb}$$

(c) (i) As the coil rotates, the component of the magnetic flux density perpendicular to the surface changes, hence the magnetic flux linkage changes.

(ii) Average e.m.f.

$$= \frac{\Delta\phi}{t} = \frac{(0.625 - 0)}{0.25 \times 10^{-3}} \\ = 2500 \text{ V}$$

Note: Do take note of the milli prefix in question.

(iii) The maximum e.m.f. is the gradient at the steepest parts of the graph. According to Faraday's law, induced e.m.f is directly proportional to rate of change of magnetic flux. Hence the ,maximum e.m.f corresponds to greatest change in magnetic flux linkage.